

Expanded Lineup of lower-power 32 bit MCU for portable and sensing applications



The RX100 series is the RX Family's new entry level 32-bit MCU, extending the RX portfolio to the low end of the spectrum in terms of pin count and flash memory size. It delivers 1.56DMIPS/MHz, and 70uA/DMIPS, consumes only 350nA in sleep mode and snaps into full operation in just 4.8µs, is the kind of power consumption typically only found in the 8-bit MCUs.

The new entrant is a great fit for those who want to benefit from the higher performance RX 32-bit architecture at the lowest possible cost.



Expanding RX family

Realize both low end high performance

- 32bit product lineup with small-pin count/small-size Flash
- High performance, high functions & cost effectiveness (3.08CoreMark/MHz)
- Fast wake-up (4.8µs) from standby mode to run mode

Shorten development TAT

- Compatibility between RX200 & 600, scalability
- Common development environment among RX series

Pursue lower power consumption

- World's lowest current performance ratio (14.9CoreMark/mA)
- Low current consumption (0.1mA/MHz) in run mode
- Low standby current (0.35µA) without power shutdown

RX100 features

Low Power, Fast Wakeup

- 100µA/MHz* (* All peripherals OFF, running NOP.)
- 350nA standby, 4.8µs wake-up

USB Connectivity

- Host, device and OTG

Safety features

- Built in safety features (CAC, DOC, I-WDT, GPIO)
- Temperature sensor

Scalable

- Fully compatible with RX600 and RX200
- Low Pin Count (36-64 pins), 16KB to 128KB
- Multifunction Pin Controller (MPC)

Zero wait-state Flash

- 1KB Block size
- Erase/Write operation down to 1.8V
- BGO Data Flash (Programmable while code is executed)

RX100 Tools

The Renesas Starter Kit for RX111 is intended as a user-friendly introductory and evaluation tool for the RX111 microcontroller. Find out more from

<http://sg.renesas.com/rx111rsk>

